D Data
S Structures
A Algorithms

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Data Structures & Algorithms

Today's Topic: Graph Algorithms

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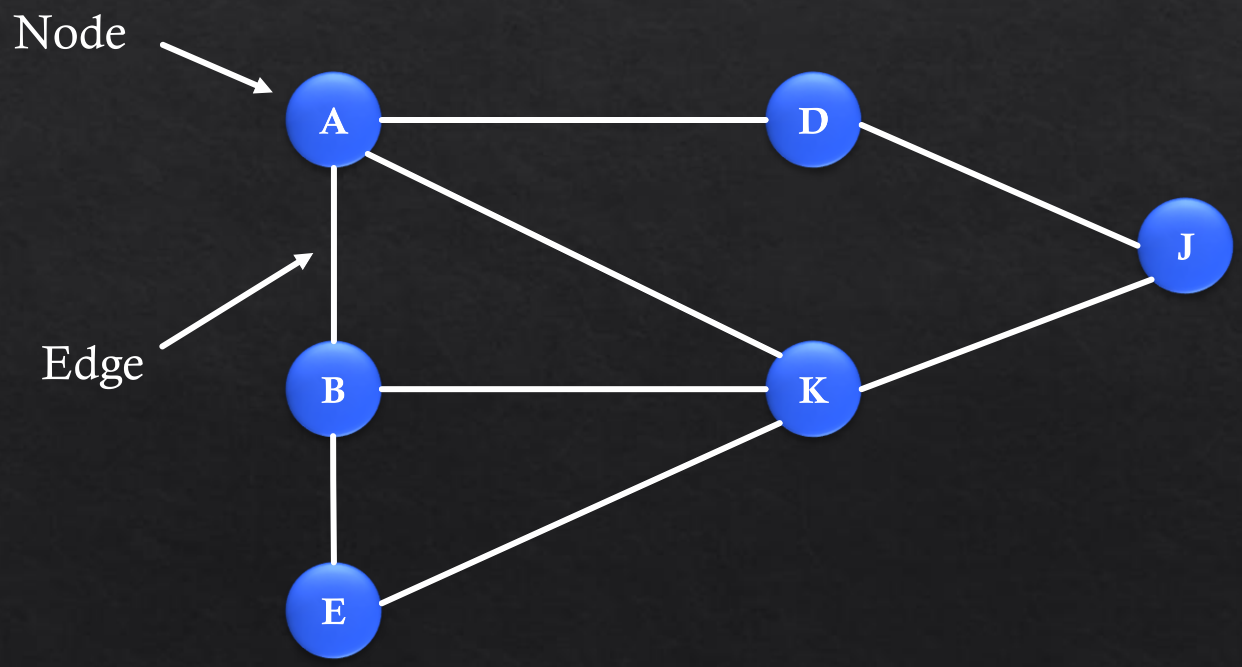
What is a Graph?

- A graph is a way of representing relationships that exist between pairs of objects.
- That is, a graph is a set of objects, called vertices, together with a collection of pairwise connections between them, called edges.



What is a Graph?

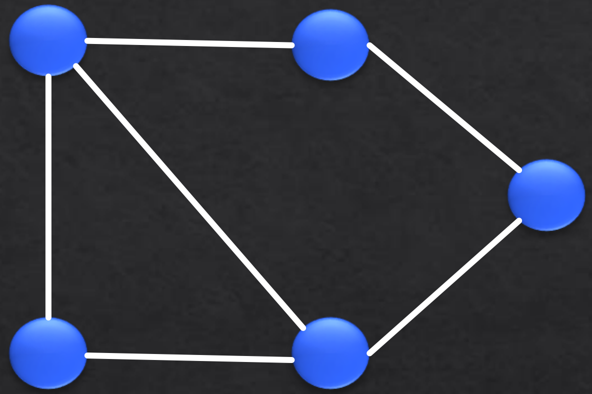


Image Source: en.wikipedia.org/wiki/Geography_of_the_United_States

Graph Terminology

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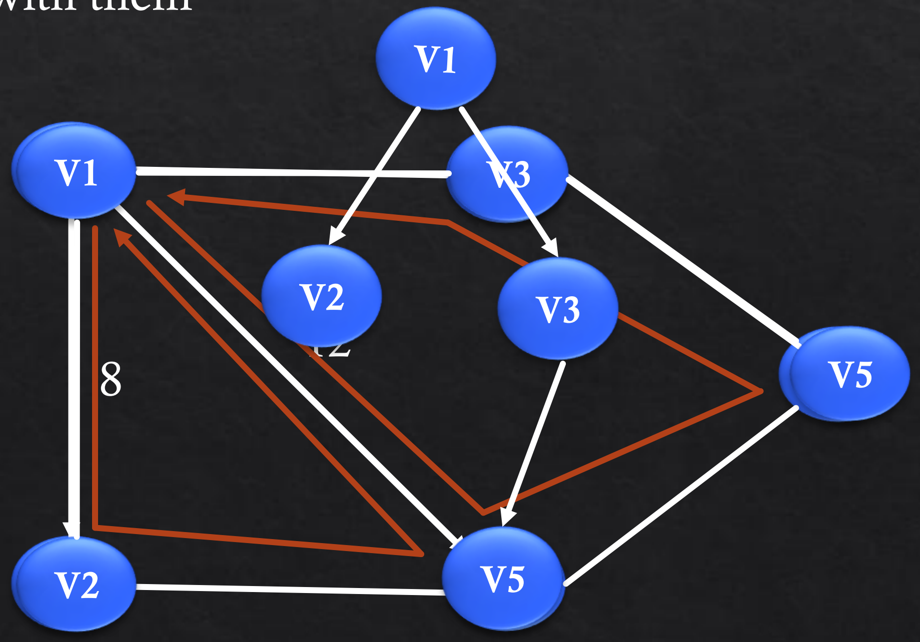
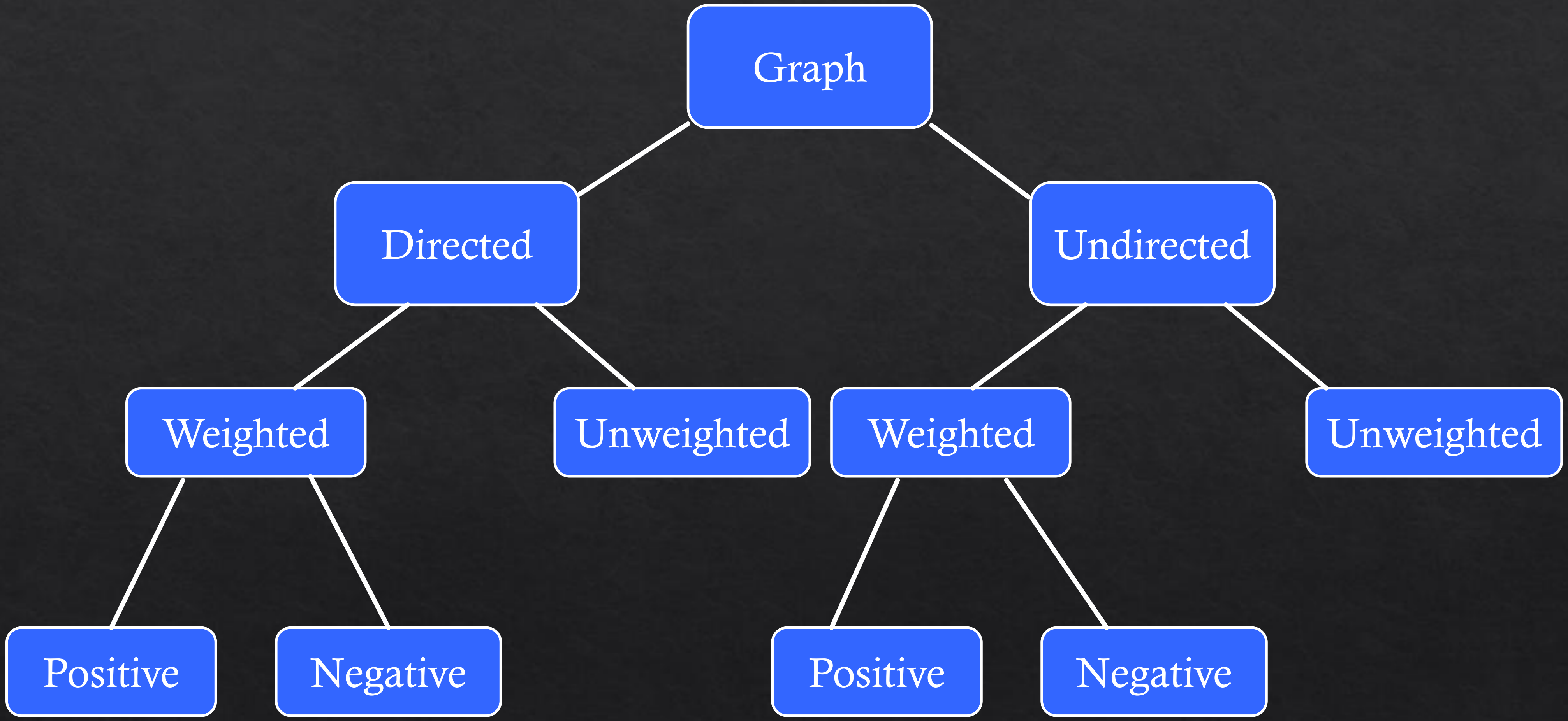
- Vertices (vertex): Vertices are the nodes of the graph
- Edge: The edge is the line that connects pairs of vertices
- Unweighted graph: A graph that does not have a weight associated with any edge
- Weighted graph: A graph that has a weight associated with any edge
- Undirected graph: In case the edges of the graph do not have a direction associated with them
- Directed graph: If the edges in a graph have a direction associated with them
- Cyclic graph: A graph that has at least one loop
- Acyclic graph: A graph with no loops
- Tree: It is a special case of directed acyclic graphs



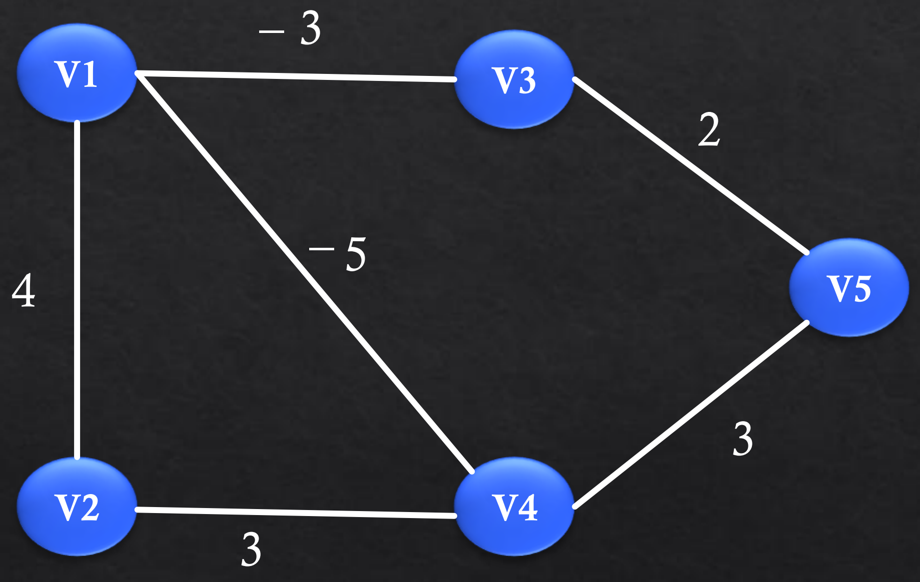
Graph Types

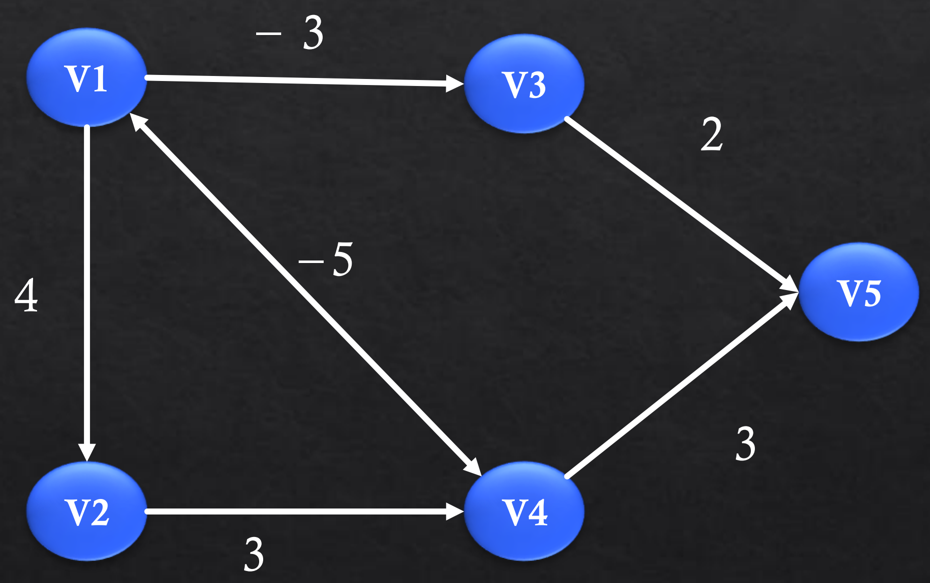

Graph
Directed
Undirected
Weighted
Unweighted
Weighted
Unweighted
Positive
Negative
Positive
Negative

Graph Types

1. Unweighted - undirected
3. Positive - weighted - undirected
5. Negative - weighted - undirected

2. Unweighted - directed
4. Positive - weighted - directed
6. Negative - weighted - directed





Graph Representation

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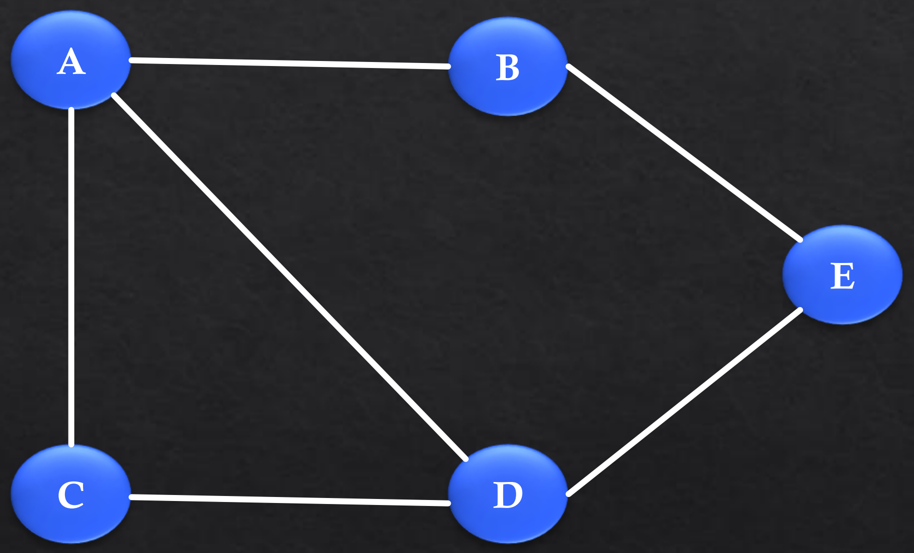
Adjacency matrix: An adjacency matrix is a square matrix, or it is a 2D array. And the elements of the matrix indicate whether pairs of vertices are adjacent or not in the graph.



	A	B	C	D	E
A	0	1	1	1	0
B	1	0	0	0	1
C	1	0	0	1	0
D	1	0	1	0	1
E	0	1	0	1	0

Graph Representation

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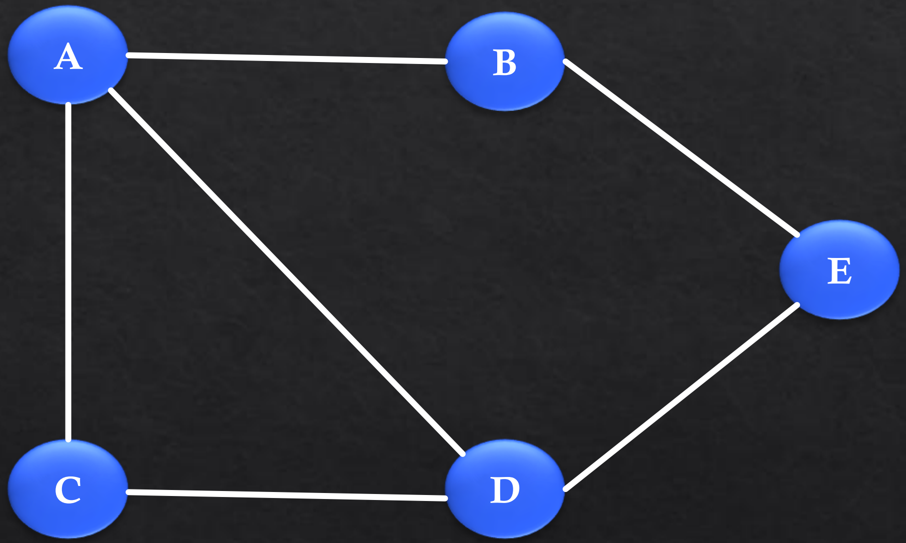
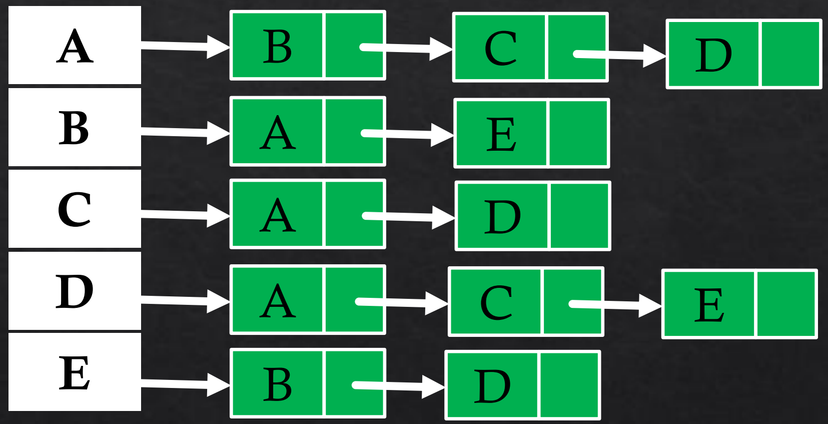
Adjacency list: An adjacency list is a collection of unordered lists used to represent a graph. Each list describes the set of neighbors of a vertex in the graph.

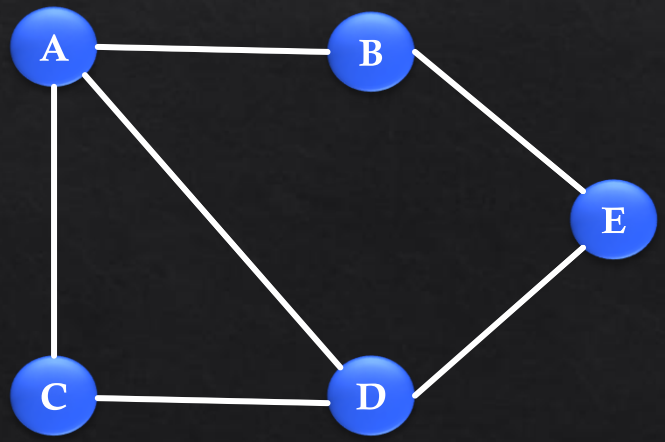


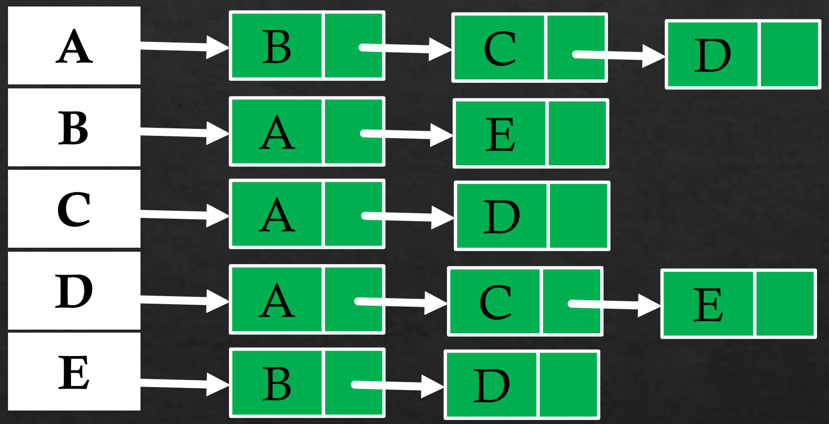


Graph Representation

- If a graph is complete or nearly complete, we should use an Adjacency Matrix.
- If the number of edges is few, then we should use an Adjacency List.

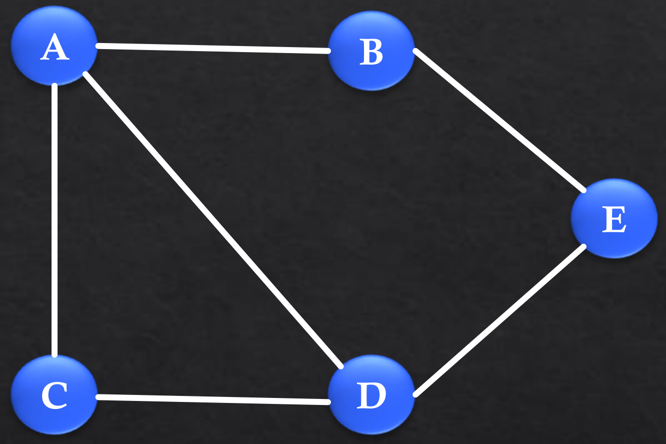
	A	B	C	D	E
A	0	1	1	1	0
B	1	0	0	0	1
C	1	0	0	1	0
D	1	0	1	0	1
E	0	1	0	1	0





Graph Representation

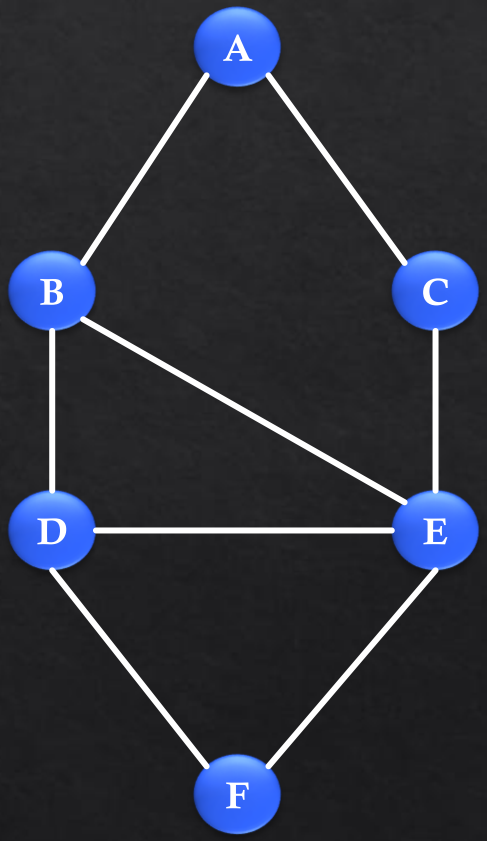
- Python Dictionary implementation




```
{
    A: [B, C, D],
    B: [A, E],
    C: [A, D],
    D: [A, C, E],
    E: [B, D]
}
```

Graph Representation

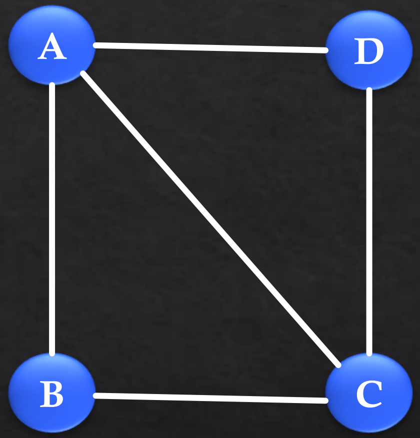
- Python Dictionary implementation




```
{
    A: [B, C],
    B: [A, D, E],
    C: [A, E],
    D: [B, E, F],
    E: [C, D, F],
    F: [D, E]
}
```

Graph Representation

- Python Dictionary implementation




```
{
    A: [B, C, D],
    B: [A, C],
    C: [A, B, D],
    D: [A, C]
}
```